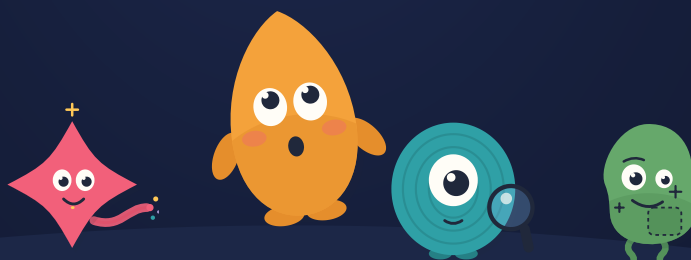


The First Book *artificial* of Intelligence

A Children's Journey Into How Minds Learn, Think and Create

FOR ABSOLUTE BEGINNERS · BOOK ONE OF THREE





**Turns out writing a book is easier than
convincing someone to publish it.**

So I skipped that part.

I wrote these books for my favourite people. Since
you're here, I'm assuming that's you.

They're free until a publisher discovers my existence,
offers me a contract, and starts sending royalty
cheques.

Enjoy the book. If you like it, drop me a note. If you
love it—or if it sparks a new idea for someone you
love—Hot Wheels and LEGO remain valid currency
in my kingdom.



DIO • STESSO

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The First Book *artificial* of Intelligence

A Children's Journey Into How Minds
Learn, Think and Create



BY DIO.STESSO
code.dio.stesso@gmail.com

The Journey

Five lands · twenty-five ideas · one big discovery.



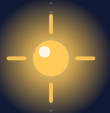
How Minds Learn

prediction · inference · memory · learning · fixing mistakes



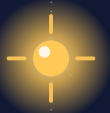
How Machines Learn

pattern · features · similarity · data · training



How Machines Notice Things

model · sorting · deciding · unfairness (bias) · network



How Machines Make Things

deep learning · word machines · making new things · confident mistakes · checking



What Makes Us Special

fairness · good judgment · wisdom · working together · intelligence

...and the big discovery: it was all inside you.

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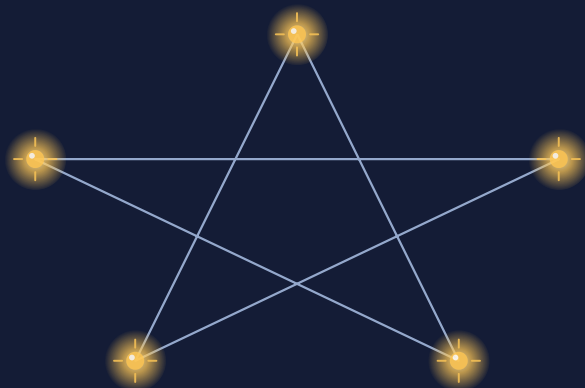
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The Hill and the First Spark



One night, Pixel
found a tiny light
that no one had ever noticed.

Pixel touched it —
and a little path
lit up, leading into the dark.

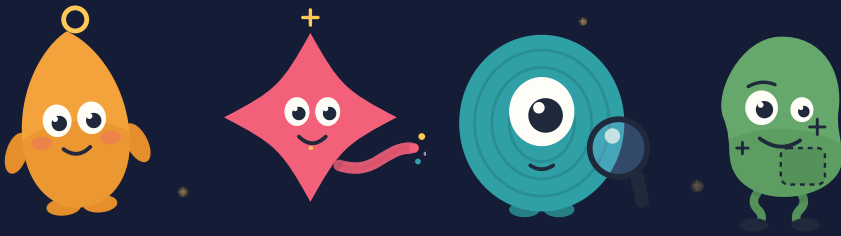


That little light was the very first *spark* of an idea. Pixel did not know it yet — but a big journey was about to begin. A journey to find out how minds think, and learn, and make new things. Come along, and watch the sparks add up!



PART ONE

How Minds Learn



First, let's see how a mind
begins to know things.

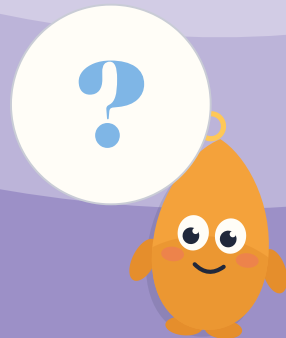
The Wrapped Present



Nova has a present for Pixel.
It is wrapped up tight. You
cannot see inside.



But Pixel is smiling. “I think I know what it is!” says Pixel. How can Pixel know? The present is still closed!



LET'S FIGURE IT OUT



Pixel gives the present a little shake. Rattle, rattle! Pixel remembers that toy blocks make that sound. So Pixel makes a smart guess: maybe it is blocks.

AND HERE'S THE TRICK



Then Pixel feels the box. It is long and thin. A ball is round — so it is NOT a ball. Maybe it is a paintbrush! Each clue helps Pixel know a little more.



HERE'S THE SECRET

When you use clues and what you already know to figure out what you cannot see yet, you are making a good guess. The more clues you have, the better your guess.



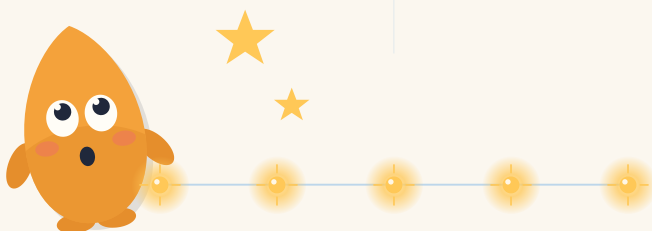
*Think about it: What makes a guess a GOOD guess?
(Hint: a good guess has a reason behind it.)*

Grown-ups have a special word for a guess about what comes next.



PREDICTION

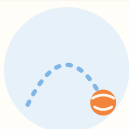
You make these every day!



You do this too —



when you sing the next word of a song you know



when you put your hands where a ball will land



when your pet sits by its bowl and you know it's hungry

AND THE MACHINES?

One day you will learn that machines guess what comes next too. They use clues, just like you.

TRY IT YOURSELF

Sing a song to a friend, but stop right before the last word. Can they guess it? Ask how they knew!

FOR GROWN-UPS

Predict together: "What do you think is for dinner?" Then find out. Ask what clues helped.

The Wet Umbrella



Echo comes inside and
stops. The sun is out.
The sky is blue. But the
umbrella is dripping wet!



Nobody told Echo what happened. So how can Echo know that it rained? Echo did not even see the rain come down!



LET'S FIGURE IT OUT



Echo looks at the clues. The umbrella is wet. There is a little puddle below it. Wet umbrellas come from rain. So Echo knows: it must have rained a while ago.

AND HERE'S THE TRICK



Then Echo spots muddy paw-prints by the door. The cat is asleep — with muddy feet! Echo never saw the cat go out. But the prints tell the whole story.



HERE'S THE SECRET

**Clues that are left behind
can tell you about things
you did NOT see happen.
When you read the clues,
you become a detective.**



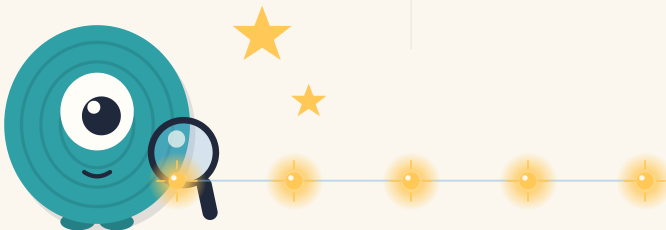
*Think about it: What new clue would make you change
your mind?*

There is a word for figuring out the unseen from clues.



INFERENCE

You are a clue-reader!



You do this too —



crumbs on a plate — someone had a snack



a door left open — someone went outside



a warm cup — someone was just here

AND THE MACHINES?

Later, machines read clues too — in pictures and words — to figure out what they cannot see.

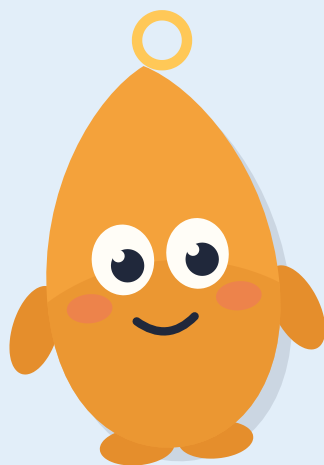
TRY IT YOURSELF

Be a detective! Find one clue at home and guess what happened. “The towel is wet, so...”

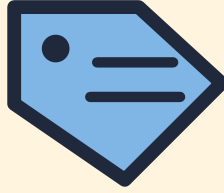
FOR GROWN-UPS

Play detective together: “What do you think happened here?” Let the clues do the talking.

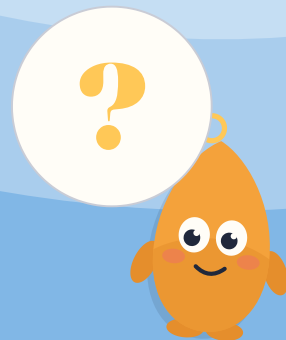
The River of Stones



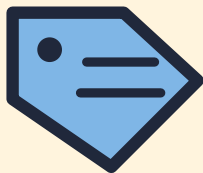
Pixel walks beside a magic river. Each time Pixel remembers something, the river gives a glowing stone.



Pixel holds a stone for a friend's name, and one for the way home. But how do these little stones help Pixel? They are only memories!



LET'S FIGURE IT OUT



When Pixel meets the friend again, Pixel looks at the name-stone and says, “Hello, Sam!” The stone kept the name safe, so Pixel could use it again.

AND HERE'S THE TRICK

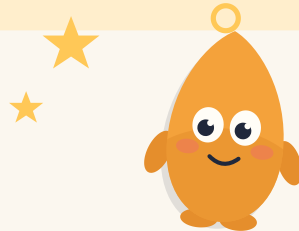


When it is time to go, Pixel
follows the way-home stone.
Pixel walked that path so often
that the memory stayed bright.
Remembering the past helps
Pixel right now.



HERE'S THE SECRET

Your memory keeps things from the past, so you can use them today. What you remember helps you think and helps you find your way.



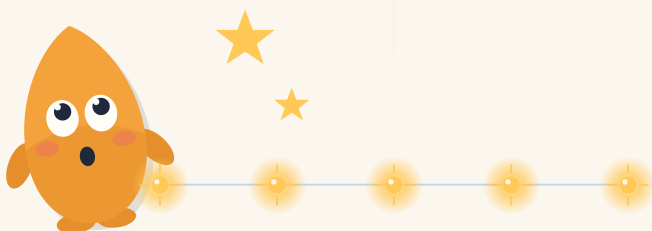
Think about it: Why do we forget some things but remember others?

There is a word for keeping things in your mind to use again.



MEMORY

Your mind is full of these!



You do this too —



you remember a song and hum it later



you remember your friends' names



you remember how a fall felt, so you're careful

AND THE MACHINES?

Soon you'll learn that machines keep things to remember too. We call what they keep *data*.

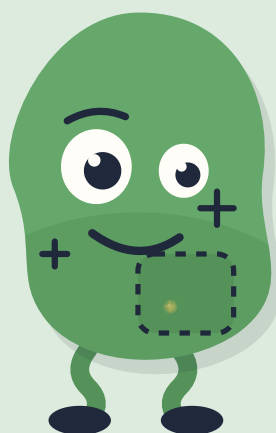
TRY IT YOURSELF

Play a memory game: look at five toys, close your eyes, and try to name them all!

FOR GROWN-UPS

Share a memory together — from a smell or a song. One little thing can bring back a whole day.

The Garden of Dogs



Milo has never
seen a dog before.
Then, one by one, lots of
dogs come to play.



A big fluffy one. A tiny smooth one. They all look so different! How will Milo ever know what a dog really is?

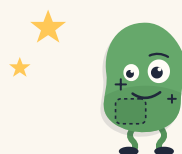


LET'S FIGURE IT OUT



Milo watches them all play.
Every dog wags its tail. Every
dog says “woof.” Every dog has
a wet nose. Milo notices the
things they all share.

AND HERE'S THE TRICK

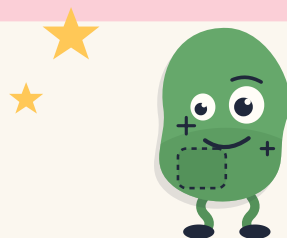


Then a cat walks by. Soft fur, like a dog — but pointy ears, and it says “meow.” “That’s NOT a dog!” says Milo. Now Milo really knows what a dog is.



HERE'S THE SECRET

**You learn what something
is by looking at lots of
examples and noticing
what they share. The more
examples you see, the
better you know.**



*Think about it: How many dogs do you need to meet
before you really know what a dog is?*

There is a word for getting to know something
from many examples.



LEARNING

You do this all the time!



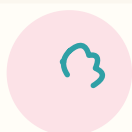
You do this too —



you learn “triangle” from lots of triangles



you learn “kind” from lots of kind acts



you learn to whistle by trying and trying

AND THE MACHINES?

In Part Two, you’ll see that machines learn from examples too. People call it *machine learning*!

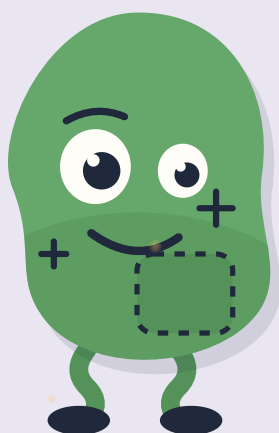
TRY IT YOURSELF

Find five round things in your home. What do they all share? What makes them round?

FOR GROWN-UPS

Point at things and ask “why is that a dog?” Naming what they share is the whole lesson.

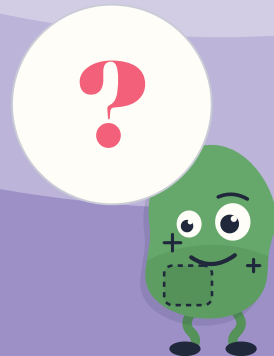
The Mountain Path



Milo is climbing a mountain.
Milo takes a turn —
oops, the wrong way! Milo
has to go back.



Milo feels a little sad about the mistake. But then Milo sees a flower growing right where he slipped last time. Can a mistake be a GOOD thing?



LET'S FIGURE IT OUT



Milo thinks about learning to ride a bike. Wobble, wobble, fall.

But each fall taught Milo something new: lean here, push there. Soon — Milo could ride!

AND HERE'S THE TRICK

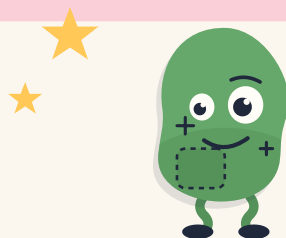


Milo remembers a cake that came out too salty. Yuck! So next time, Milo used less salt. The mistake showed Milo exactly how to make it better.



HERE'S THE SECRET

A mistake is not a bad thing. A mistake shows you what to fix. When you notice it and try again, you get a little better each time.



Think about it: Is a mistake a failure... or a clue about what to do next?

There is a word for noticing a mistake and making it right.



FIXING MISTAKES

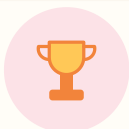
Mistakes help you grow!



You do this too —



a wrong note shows you the right one



losing a game helps you find a better plan



saying a word wrong helps you learn to say it right

AND THE MACHINES?

Later you'll see that machines get better by fixing their mistakes, over and over. It's called *training*.

TRY IT YOURSELF

Try something tricky — like stacking blocks really high. When they fall, what will you change next time?

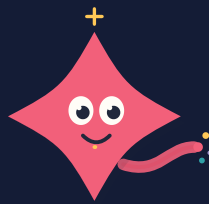
FOR GROWN-UPS

When a slip happens, ask “what will we try next time?” instead of “whose fault was it?”

The background of the entire page is a dark navy blue sky filled with numerous small, golden-yellow stars of varying sizes. At the bottom, there are three horizontal, wavy bands of color representing a landscape: a light blue band at the top, a medium teal band in the middle, and a light green band at the bottom. The text is centered horizontally and vertically in the upper half of the image.

PART TWO

How Machines Learn

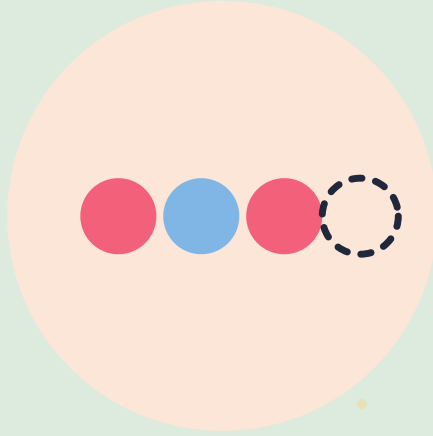


Now — could a machine learn
in the same way we do?

The Painted Floor



Echo finds a floor
of colored tiles.
Red, blue, red, blue, red,
blue...



Echo's eyes light up.
“I know what color is
next!” says Echo. But the
next tile is hidden. How
can Echo be so sure?

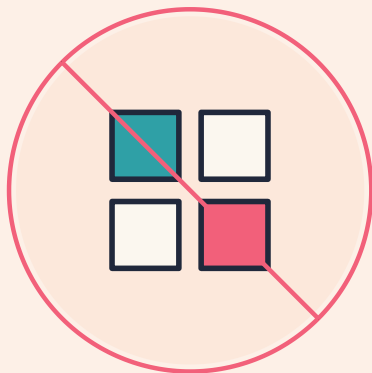


LET'S FIGURE IT OUT



Echo looks at the order. Red, blue, red, blue. It keeps repeating! After blue comes red again. So the next tile must be red. Echo is right!

AND HERE'S THE TRICK



Then Echo finds a tile that is wrong — a green one where a red should be. Echo spots it right away, because it breaks the repeating pattern.



HERE'S THE SECRET

A pattern is something that repeats. When you see HOW it repeats, you can tell what comes next — or spot when something doesn't fit.



Think about it: Where else do you see patterns? In music? On your clothes?

There is a word for something that repeats in order.



PATTERN

Patterns are everywhere!



You do this too —



the days go in the same order every week



a drumbeat repeats, so you can clap along



2, 4, 6... you can tell what comes next

AND THE MACHINES?

Machines are very good at finding patterns — even in huge piles of things. You'll see how!

TRY IT YOURSELF

Make a clap pattern: clap-clap-stomp, clap-clap-stomp. Can a friend keep it going?

FOR GROWN-UPS

Spot patterns together — on tiles, fences, and songs. A pattern is a tiny prediction you can see.

Apples and Oranges



Echo holds an apple
and an orange.

“Tell them apart — but you
can’t use color!”



Without color, how can
you tell which is which?
They are both round. They
both fit in your hand.



LET'S FIGURE IT OUT



So you close your eyes and feel.
The apple is smooth and cool.
The orange is bumpy. Just by
touching, you can tell which one
is which!

AND HERE'S THE TRICK



You can use your nose, too. The orange smells zingy and fresh. You didn't need color at all — you used other clues.



HERE'S THE SECRET

**To tell things apart, you
pick a few important
things to notice — like
shape, smell, or sound.
You don't have to notice
everything at once.**



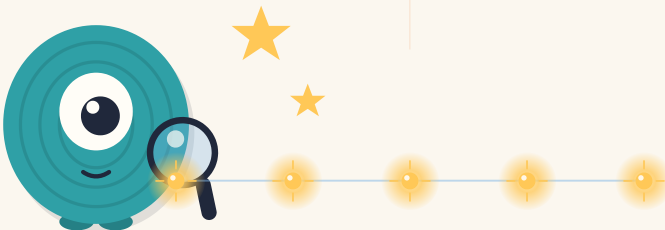
*Think about it: Which clue is the MOST helpful for telling
two things apart?*

There is a word for the special things you choose
to notice.



FEATURES

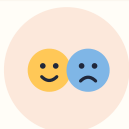
You pick these all the time!



You do this too —



floppy ears and a woof? that's a dog



a smile or a frown shows how someone feels



you can tell two leaves apart by their little lines

AND THE MACHINES?

Machines pick important things to notice too — but sometimes they pick the wrong ones. More soon!

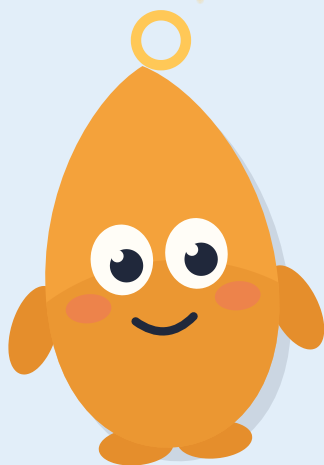
TRY IT YOURSELF

Sort your toys without using color. What did you use instead — size? shape? Can a friend guess your rule?

FOR GROWN-UPS

Play “guess my rule”: sort by a secret feature and let them figure out which one you used.

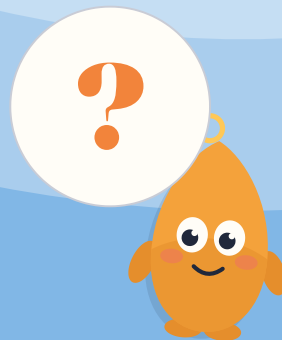
The Shells on the Shore



Pixel pours a pile of shells onto the sand.
“Let’s sort them! Which ones go together?”



But wait — should they go together by size? By color? By shape? The shells could be sorted so many different ways!



LET'S FIGURE IT OUT



big ones

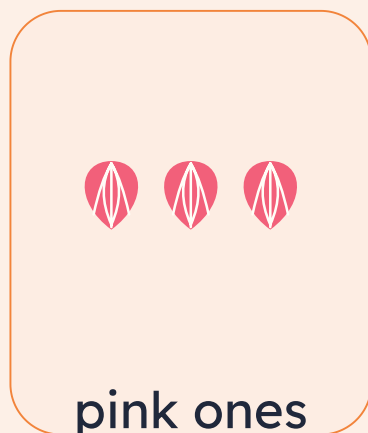
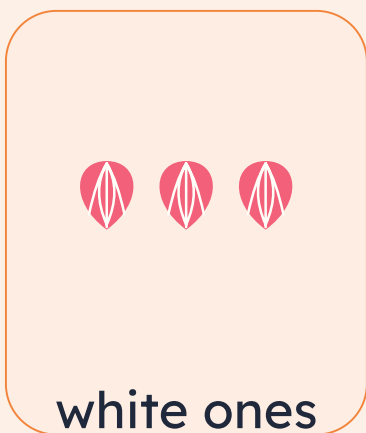


small ones



Pixel sorts by size first. All the big shells in one pile, the small ones in another. Now the big ones are “together.”

AND HERE'S THE TRICK

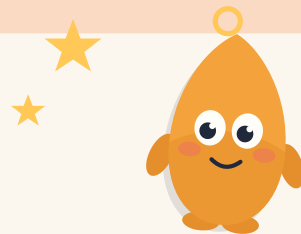


Then Pixel sorts by color instead.
The piles are different now! A
big white shell and a small white
shell are together this time.
What counts as “together”
changed!



HERE'S THE SECRET

Two things are alike when they share what you are looking at. Change what you look at, and what is “alike” changes too.



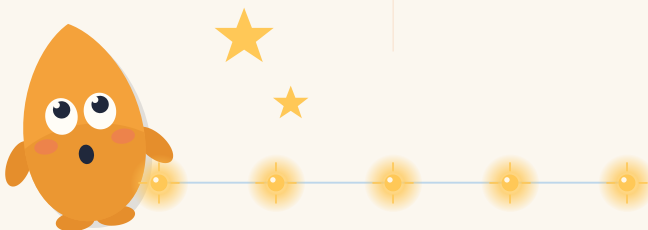
Think about it: Can two things be alike AND different at the same time?

There is a word for how alike two things are.



SIMILARITY

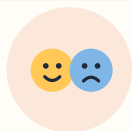
You notice this every day!



You do this too —



two socks match — same size,
same color



people in a family can look alike



two songs can feel alike, even if
they're different

AND THE MACHINES?

Machines measure how alike things are, to help
them decide what something might be.

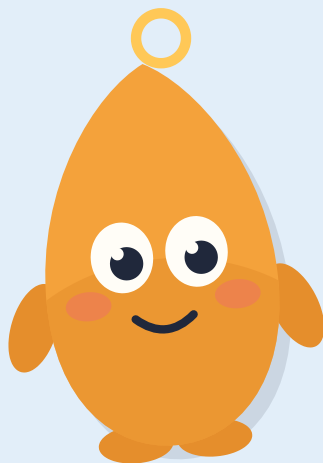
TRY IT YOURSELF

Sort your crayons by “alike.” Then change the
rule — and watch the groups change!

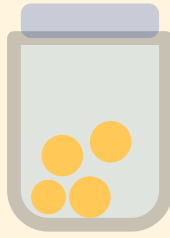
FOR GROWN-UPS

Sort buttons or socks two different ways. “Alike” depends
on the rule — that’s the big idea.

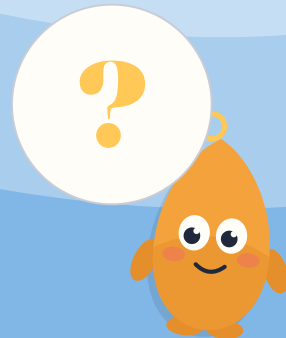
The Jar of Stones



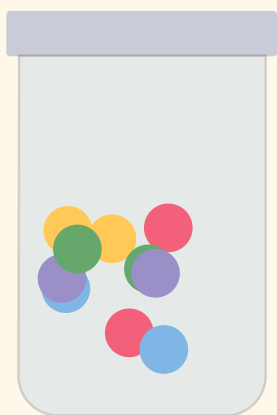
**Pixel is filling a jar
with glowing stones.
Stones from near, stones
from far, every color.**



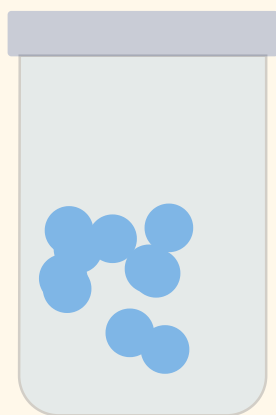
“Why so many?” asks Milo. “Why not just grab a few?” Pixel smiles. “Watch what a full jar can do!”



LET'S FIGURE IT OUT



all kinds ✓

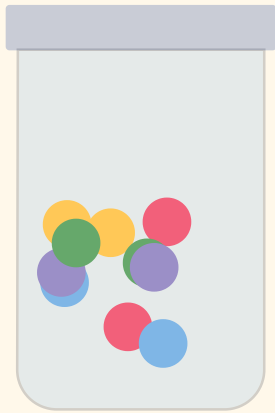


only blue X

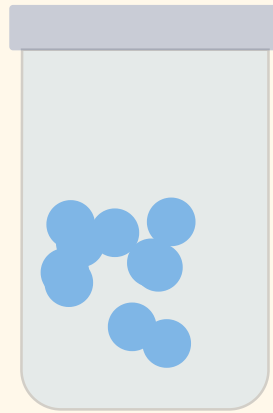


With lots of different stones, the jar can teach Pixel about the whole world — red ones, blue ones, big and small. Lots of examples means Pixel learns more.

AND HERE'S THE TRICK



all kinds ✓



only blue ✗

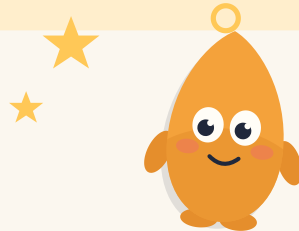


But Echo brings a jar with stones from just one street. They are ALL blue! This jar would teach Pixel that every stone is blue — and that simply isn't true.



HERE'S THE SECRET

When you gather lots of examples, you have data. And the more DIFFERENT your examples are, the more you can learn that is really true.

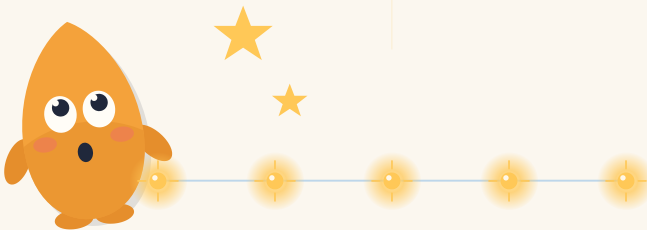


Think about it: Which teaches more: many of the SAME thing, or many DIFFERENT things?

There is a word for a big collection of examples.

DATA

You collect this too!



You do this too —



a weather chart is data about the sky



a list of who likes which fruit is data



a photo album is data about your days

AND THE MACHINES?

Machines learn from mountains of data. But if the data is lopsided, they learn the wrong thing!

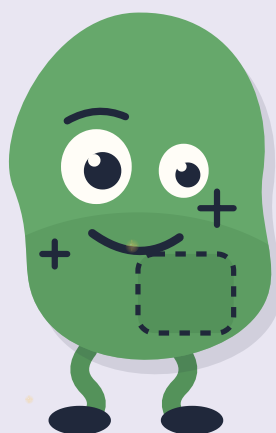
TRY IT YOURSELF

Count something for a few days — like birds at a window. Your marks become data!

FOR GROWN-UPS

Collect simple data together, and talk about how variety changes the story it tells.

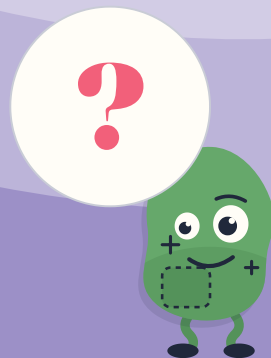
Teaching the Puppy



Milo has a new puppy and
wants to teach it to sit.
“Sit!” says Milo. The puppy
just wags its tail.



The puppy doesn't know how to sit — yet. How can Milo teach it? You can't just TELL a puppy what to do!



LET'S FIGURE IT OUT



So Milo shows the puppy. When the puppy sits, Milo gives a treat and says “Good!” The puppy starts to learn: sitting makes good things happen.

AND HERE'S THE TRICK

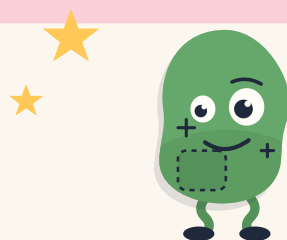


Milo does it again and again.
Try, treat, try, treat. Each time,
the puppy gets a little better.
Soon, the puppy sits the
moment Milo asks!



HERE'S THE SECRET

**You get good at
something by trying,
checking how it went,
fixing it, and trying again
— lots and lots of times.
All of that is called
practice.**



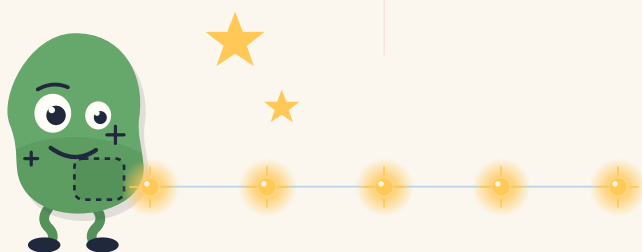
*Think about it: Could you get really good at something
without ever making a single mistake?*

There is a word for practicing over and over to
get good.

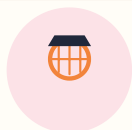


TRAINING

You do this to learn, too!



You do this too —



you practice shooting until you score



you practice a word until you can spell it



you practice a song until it sounds right

AND THE MACHINES?

When a machine practices like this, we call it *training*. It's how machines learn — coming up next!

TRY IT YOURSELF

Pick one small skill. Practice it a little each day. Watch yourself get better!

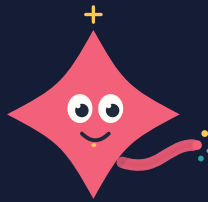
FOR GROWN-UPS

Praise the trying, not just the winning. Each small fix is real progress.



PART THREE

How Machines Notice Things



After lots of practice,
something clever takes shape.

The Paper Map



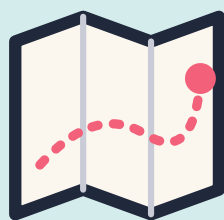
**Pixel is lost in a big town.
Then Pixel pulls out a small
paper map.**



The map has no trees, no people, no smell of bread. It is so much simpler than the real town. So how can a little flat map help Pixel find the way?



LET'S FIGURE IT OUT



The map keeps the important things — the roads, and where they go. It leaves out the rest. With just the roads, Pixel can find the way home!

AND HERE'S THE TRICK

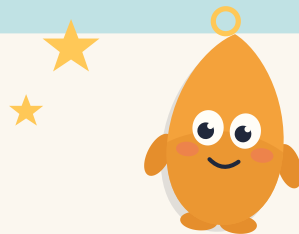


A globe is like this too. It's a tiny ball, not the whole giant Earth.
But it keeps what matters —
where the land and seas are.



HERE'S THE SECRET

**A model is a simpler copy
of something real. It
keeps the important parts
and leaves out the rest, so
it's easy to use.**



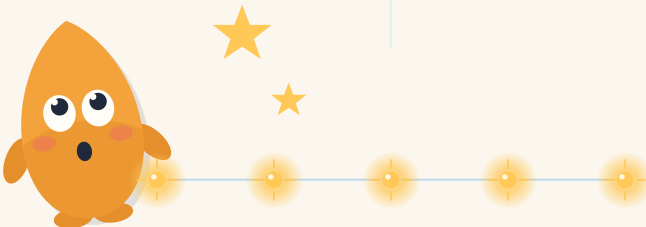
Think about it: What did the map leave out? Could leaving that out ever cause a problem?

There is a word for a simple copy that helps you understand something big.



MODEL

You use these all the time!



You do this too —



a toy car shows how a real car moves



a globe is a little model of the Earth



a recipe is a model of a cake — the plan, not the taste

AND THE MACHINES?

When a machine learns from data, it builds a *model* of the world inside itself.

TRY IT YOURSELF

Draw a map of your room. What will you keep?
What will you leave out?

FOR GROWN-UPS

Compare a map to the real street. Talk about what was kept, and why.

The Sticker Cards



Pixel has a big pile
of sticker cards.

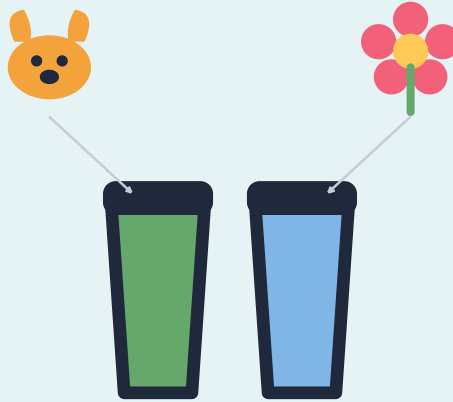
“Let’s make groups. Animals
here, plants there.”



**Pixel is sorting fast —
until one card stops her.
A mushroom! It is not
an animal. But is it a
plant? Where does it go?**



LET'S FIGURE IT OUT



Pixel looks at each card and decides its group. A dog has fur and barks — that goes with animals. A rose grows in the soil — that goes with plants.

AND HERE'S THE TRICK

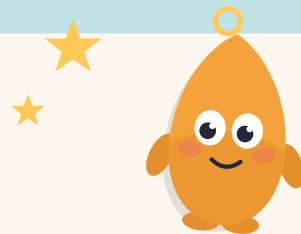


But some cards are tricky. Is a tomato a fruit or a vegetable? People put it in different groups! The groups are something WE decide.



HERE'S THE SECRET

Sorting into groups means looking at what things are like and deciding where they belong. We make up the groups ourselves, to help us.



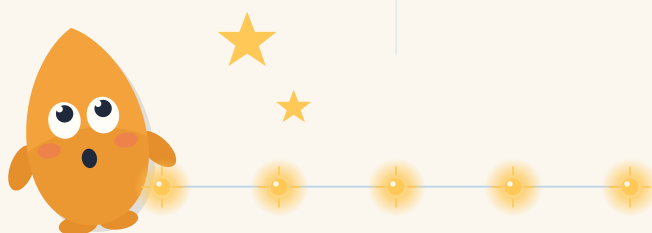
Think about it: Who gets to decide which group something belongs in?

There is a word for sorting things into groups.



SORTING

You do this every day!



You do this too —



you sort toys: blocks here, soft toys there



you sort trash: paper here, cans there



a library sorts books so you can find them

AND THE MACHINES?

Machines sort things into groups too — like “cat” or “dog.” But sometimes they sort wrong, and feel SO sure!

TRY IT YOURSELF

Sort a drawer two different ways. Find a thing that’s hard to sort. Why is it tricky?

FOR GROWN-UPS

Let tricky cases spark talk: “why is this one hard to sort?”
Groups are choices, not rules of nature.

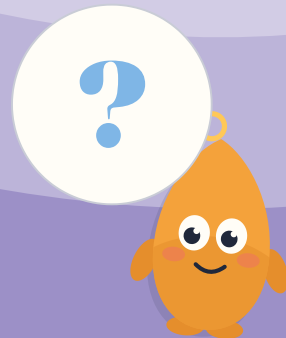
The Fork in the Path



Pixel comes to a fork in the path. One way is sunny but long. The other is short but muddy.



**Both paths are okay. So
how does Pixel choose?
There's no sign saying
which one is "right"!**



LET'S FIGURE IT OUT



Pixel thinks about what matters most. “I don’t like muddy feet,” says Pixel, “and it’s a lovely day.” So Pixel picks the sunny path.

AND HERE'S THE TRICK



Choosing is like this with a coin,
too. Spend it now on candy? Or
save it for something bigger?
You think about what you want
most.



HERE'S THE SECRET

**To make a choice, you
look at what you know,
then think about what you
care about most. Your
choice shows what
matters to you.**



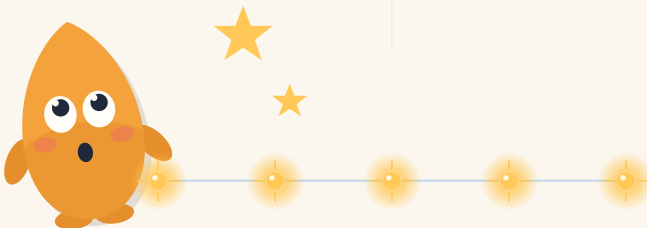
*Think about it: What did you care about most when
you chose?*

There is a word for choosing one thing out of many.



DECIDING

You make these all day!



You do this too —



you pick which book to read tonight



you decide what to wear by the weather



you choose which game to play with friends

AND THE MACHINES?

Machines make choices too, using what they learned. But **WHAT** to care about — that part comes from us.

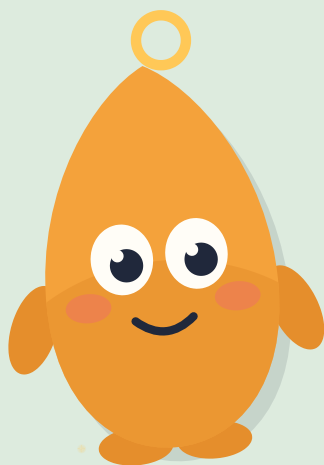
TRY IT YOURSELF

Make a choice with a chart: good things on one side, not-so-good on the other. Which side wins?

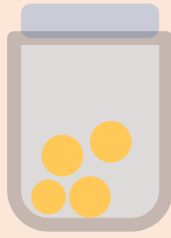
FOR GROWN-UPS

Let them choose, then ask “what made you pick that?”
Naming what we value is the lesson.

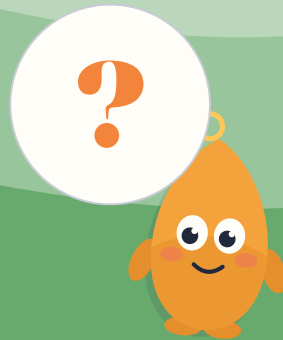
The One-Street Jar



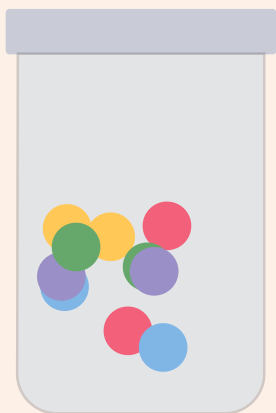
Remember Pixel's jar of stones? This one is special. Every stone inside came from just one street.



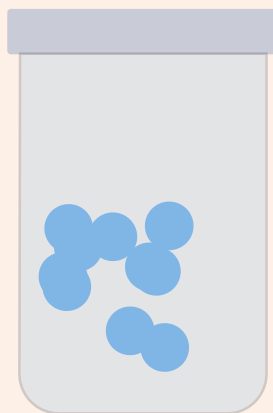
Pixel peeks in. Every single stone is blue! So the jar “thinks” that ALL stones are blue. But that can’t be true... can it?



LET'S FIGURE IT OUT



all kinds ✓



only blue ✗



Outside, the world is full of red
and green and yellow stones!
But the jar never saw those. It
only ever saw blue. So it learned
something wrong.

AND HERE'S THE TRICK

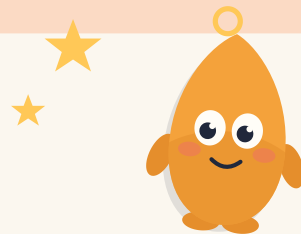


It's like meeting one grumpy dog and thinking ALL dogs are grumpy. One example is not enough — and it can really fool you.



HERE'S THE SECRET

**If your examples are
unfair, or some are
missing, then what you
learn will be unfair too.
You have to check where
your examples came from.**



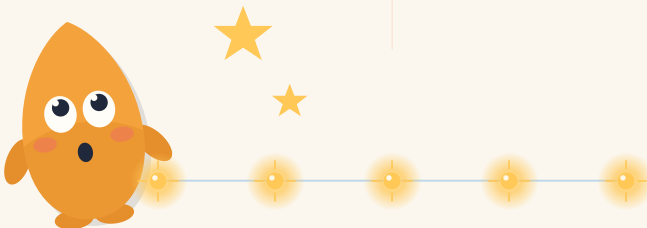
*Think about it: Whose stones are missing from the jar?
What would the jar get wrong?*

There is a word for learning the wrong thing from
unfair examples.



UNFAIRNESS (BIAS)

It's sneaky — so watch out!



You do this too —



“everyone likes this!” — but you only asked your friends



an album that skips half the family tells half the story



a vote that asks only some people isn't fair

AND THE MACHINES?

If a machine learns from lopsided data, it becomes unfair too. The fix is fairness — we'll get there!

TRY IT YOURSELF

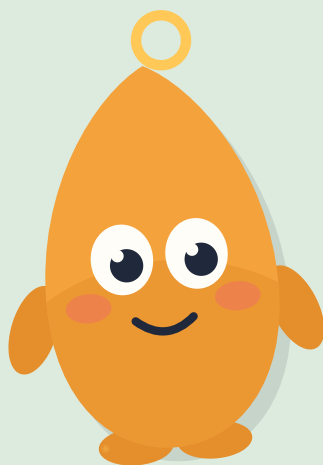
Ask one question to your family, then to some strangers. Are the answers different? Why might that be?

FOR GROWN-UPS

When a claim says “everyone,” ask together: “everyone — or just the people we asked?”

The Field of Fireflies

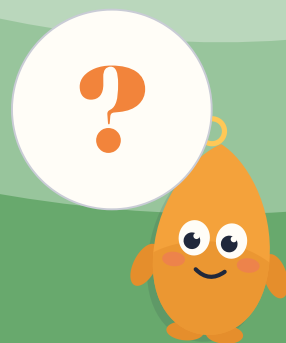
15



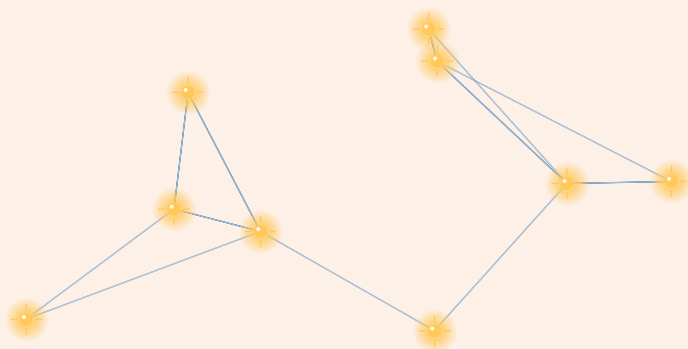
Night falls. One little
firefly blinks in the dark.
Just one tiny light — you'd
almost miss it.



**One firefly can't do much.
But then a thousand
fireflies blink together,
and the whole field
glows in great waves!
How can that happen?**

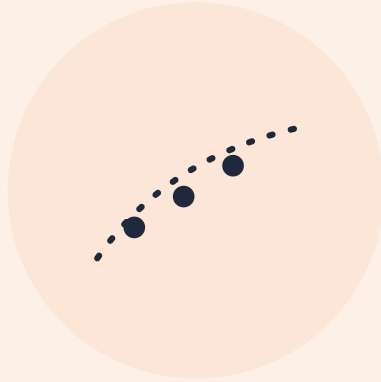


LET'S FIGURE IT OUT



Each firefly only knows one little rule: when my neighbor blinks, I blink too. One by one, they pass it along — and a big, beautiful wave appears.

AND HERE'S THE TRICK

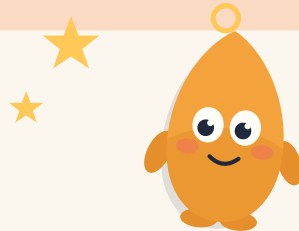


Ants are like this. No single ant knows the way to the food. But together, sharing little clues, the whole group finds the path.



HERE'S THE SECRET

**Lots of simple parts,
joined together and
sharing, can do something
clever that no single part
could do alone.**



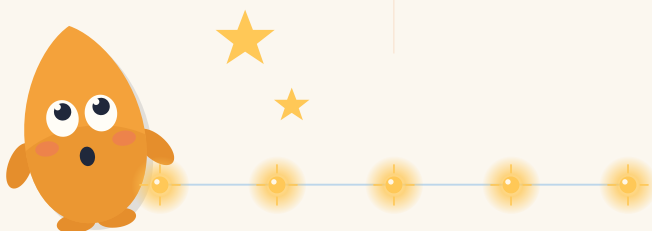
*Think about it: How can no single ant know the way,
yet the whole group find it?*

There is a word for many small parts working together, all joined up.



NETWORK

Teams work this way!



You do this too —



a line of hands passes water farther than one person



in “telephone,” a message travels mouth to mouth



many votes together pick one answer

AND THE MACHINES?

Inside a learning machine are lots of tiny parts, all joined up and working together. We call it a *network*.

TRY IT YOURSELF

Play telephone with a long sentence. Watch a message travel through many people!

FOR GROWN-UPS

Point out teams in daily life — lots of small parts making one big result.



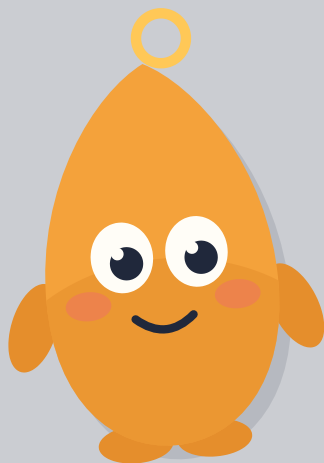
PART FOUR

How Machines Make Things



If a mind can notice...
can it also *make*?

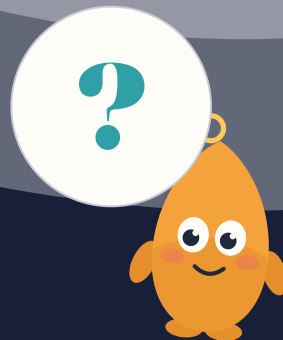
Diving Down



Pixel takes a deep breath
and dives into the sea.
Near the top it's bright.
Deep down, big surprises.



How do you understand something really big — like a whole picture, or a whole story? It feels like too much all at once!



LET'S FIGURE IT OUT

towers

walls

base



Think of a sandcastle. You don't build it all at once. First the base. Then the walls. Then the towers on top. Layer by layer, it grows.

AND HERE'S THE TRICK

story

words

letters

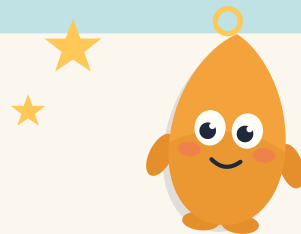


Reading is like this. First you learn letters. Then letters make words. Then words make a story. Each step is built on the one before.



HERE'S THE SECRET

**You can understand
something big by building
up in layers — simple
things first, then bigger
ideas resting on top of
them.**



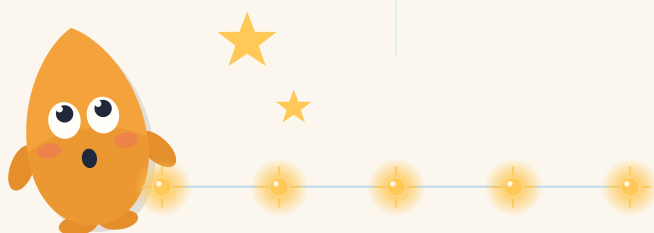
*Think about it: What do you have to learn FIRST, before
the next part can make sense?*

There is a name for learning in lots of layers, from simple to deep.



DEEP LEARNING

You learn in layers too!



You do this too —



2 4 6

first you count, then add, then multiply



first lines, then eyes, then a whole face



an onion has layer after layer, all the way down

AND THE MACHINES?

Machines can stack many, many layers to understand hard things like pictures and words. That's *deep learning*.

TRY IT YOURSELF

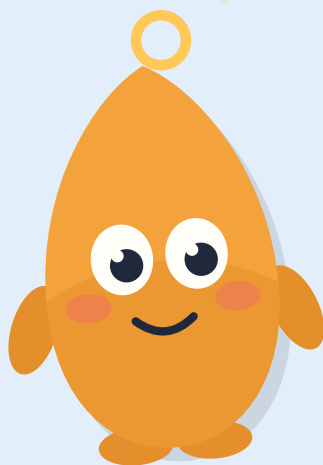
Build something in layers — blocks, or a drawing that starts with simple shapes.

FOR GROWN-UPS

Notice steps in learning: sounds before words, words before reading. Layers are everywhere.

Finish the Sentence

17



Pixel starts a sentence and stops:
“The cat sat on the...”
Everyone shouts the next word!

on the

?

**“Mat!” they all say at once.
But how did everyone
know? Pixel never told
them the ending. How
can you guess a word
you haven’t heard yet?**

?



The cat sat on the



You've heard "the cat sat on the mat" so many times. Your mind learned which words go together. So "mat" pops right up, all by itself!

Twinkle, twinkle, little

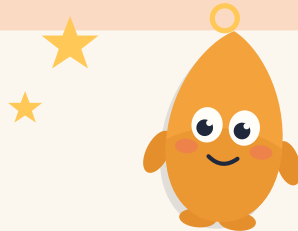


It works with rhymes, too.
“Twinkle, twinkle, little...” — and
you already want to say “star.”
You learned it from hearing it
again and again.



HERE'S THE SECRET

**When you hear words lots
and lots of times, you
learn which words tend to
come next. Then you can
finish a sentence all by
yourself.**



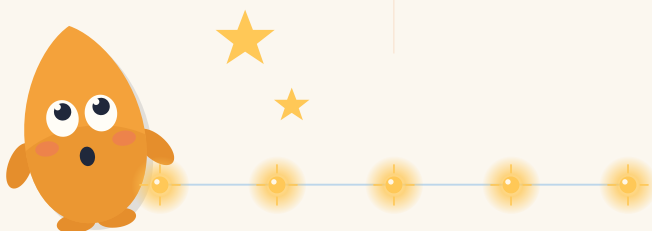
*Think about it: Is guessing the next word the SAME as
really understanding the story?*

There is a name for something that learns words
and guesses what comes next.



WORD MACHINES

You're great at this!



You do this too —



“better late than...” — you can finish it!



you guess the next word your friend will say



in a story you know, you say the lines along

AND THE MACHINES?

The chatty machines people talk to work this way! They guess the next word, over and over. We call them *language models*.

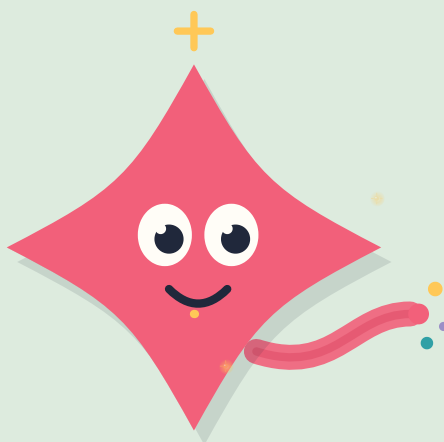
TRY IT YOURSELF

Start a sentence and let a friend finish it. Then try to surprise each other with a silly word!

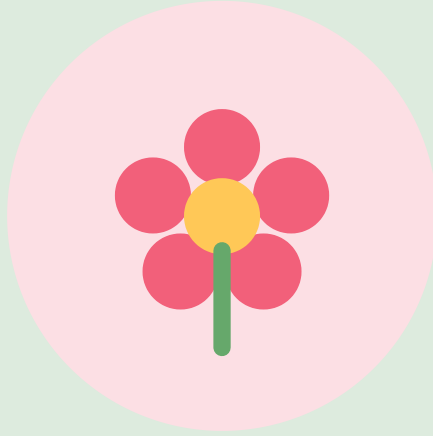
FOR GROWN-UPS

Play word tennis: each person adds one word. Notice how the words pull at each other.

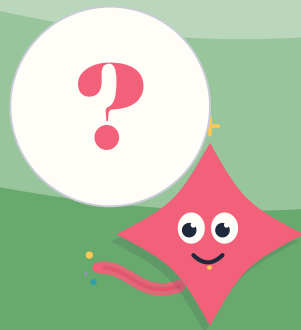
Two Flowers, One New



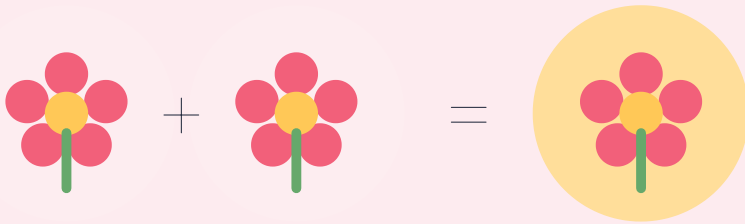
Nova loves to make things. Today, Nova takes two flowers — a red one and a yellow one — and mixes them.



Nova has never seen the
flower that will grow.
Nobody has! So how can
Nova make something
brand new out of old things?



LET'S FIGURE IT OUT



Up comes a flower with red AND yellow petals — a flower that never existed before! Nova made it by mixing two things she already knew.

AND HERE'S THE TRICK

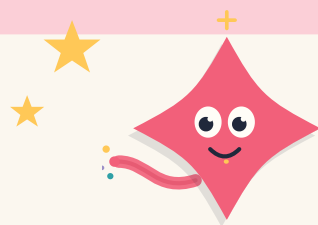


It works with music, too. You take notes you already know, and put them in a new order. Now it's a tune nobody has ever heard!



HERE'S THE SECRET

**To make something new,
you take pieces you
already know and put
them together in a new
way. New things come
from mixing old things.**



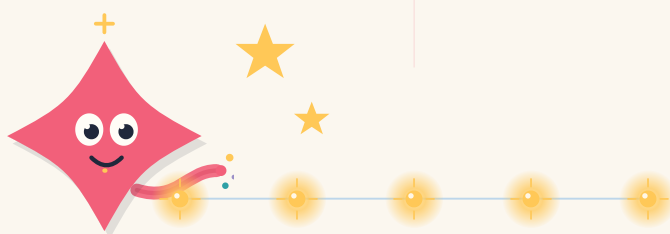
*Think about it: Is anything ever TOTALLY new — or is it
always old pieces joined in a new way?*

There is a word for making something new from
things you know.



MAKING NEW THINGS

You're an inventor!



You do this too —



mix two animals to invent a new creature



mix two ideas to tell a brand-new story



mix two snacks to make a new treat

AND THE MACHINES?

Machines can make new things too — pictures, words, and music — by mixing patterns they learned. We call it *generating*.

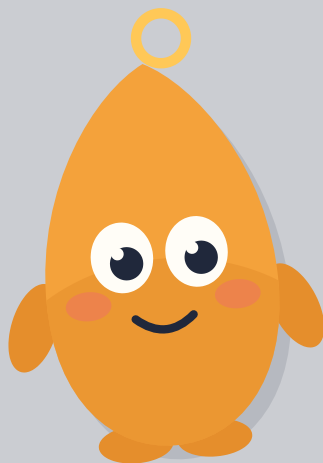
TRY IT YOURSELF

Invent an animal from two real ones. Give it a name. Draw it. You just made something new!

FOR GROWN-UPS

Make up a creature together. Notice that “new” is built from things you already knew.

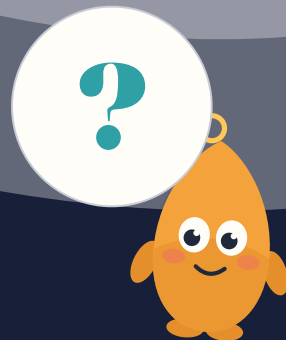
The Island That Wasn't



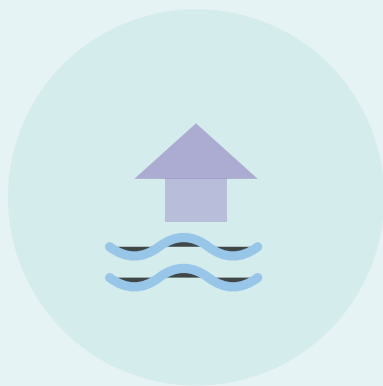
From the boat, Pixel
sees an island far away.
“Land!” shouts Pixel, sure as
can be.



But the closer they sail,
the more the island fades.
It melts into thin air! It
was never really there.
How could Pixel be so
sure — and so wrong?



LET'S FIGURE IT OUT



It was a mirage — a trick of the warm air that looks just like land. It looked so real. But looking sure is NOT the same as being true.

AND HERE'S THE TRICK

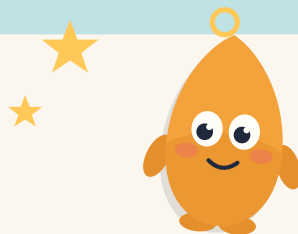


Sometimes you remember something SO clearly — and it turns out it never happened that way. Your mind can feel certain and still be wrong.



HERE'S THE SECRET

**Something can look sure,
and sound sure, and still
not be true. So when it
really matters, you check
before you believe.**



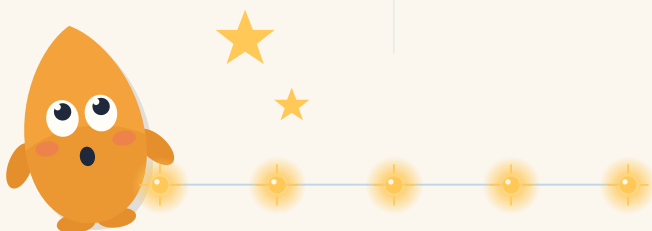
Think about it: How can you tell a SURE guess from a TRUE one?

There is a word for being very sure about something that isn't real.



CONFIDENT MISTAKES

So always check!



You do this too —



a dream can feel real until you wake up



“telephone” can turn a phrase into nonsense



a made-up fact can sound very sure

AND THE MACHINES?

Sometimes machines say things that sound sure but aren't true. It's called a *hallucination* — so we check!

TRY IT YOURSELF

Tell a tiny made-up “fact,” then look it up in a book. Did sounding sure make it true?

FOR GROWN-UPS

Model curiosity, not alarm: “that sounds amazing — how could we check it?”

Checking the Map



Echo finds an old treasure map. “X marks the spot!” But Echo doesn’t run off digging. Echo stops to think.



**“What if the map is wrong?”
says Echo. “Or even fake?”
How can Echo find out if
the map can be trusted —
before wasting a whole day?**



LET'S FIGURE IT OUT



Echo checks one little thing first.
The map shows a big tree by the
river. So Echo walks to the river
— and yes, the tree is really
there! Now the map feels more
true.

AND HERE'S THE TRICK



Then Echo looks closer with a lens and finds a corner that was drawn on later — a fake part! Checking helped Echo spot the trick.



HERE'S THE SECRET

You don't have to just believe, or just doubt. You can CHECK! Test one part against what is real. Trust grows with every check.



Think about it: What's the difference between "it sounds right" and "I checked it"?

There is a word for checking if something is really true.



CHECKING

Checking makes you strong!



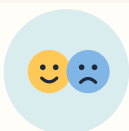
You do this too —



check a fact in two books, not just one



not sure how long? measure it!



ask a second friend who saw it too

AND THE MACHINES?

Clever machines can be wrong. So WE check their work. Being a good checker is a human superpower!

TRY IT YOURSELF

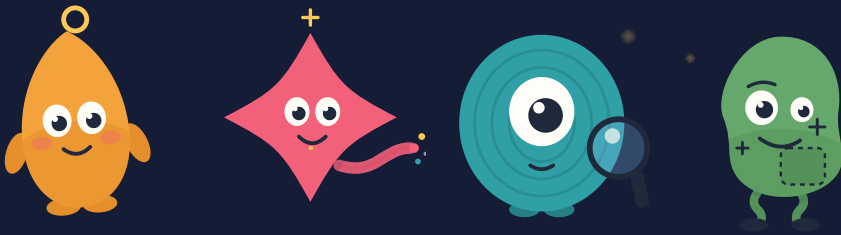
Test a claim today: “this cup holds more.” Pour and find out. Checking beats guessing!

FOR GROWN-UPS

When you hear a surprising fact, ask together: “how could we find out if it’s true?”

PART FIVE

What Makes Us Special



Machines can do so much now.
So what stays truly *ours*?

Sharing the Cake



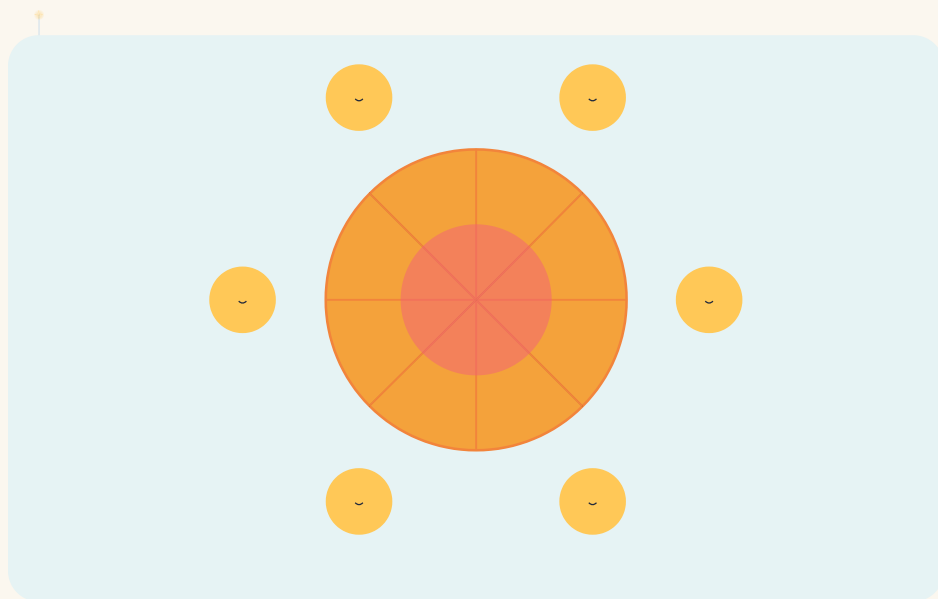
There is one cake and
lots of hungry friends.
“Let’s share it!” says Pixel.
But how, the RIGHT way?



Should everyone get the exact same slice? But what about the friend who walked a long way and is extra hungry? What is fair?



LET'S FIGURE IT OUT



Pixel cuts equal slices, so no one feels left out. Everyone is counted. That feels fair to everybody.

AND HERE'S THE TRICK

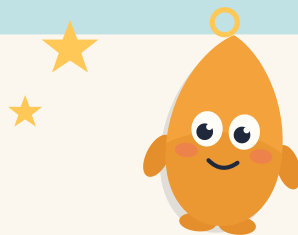


Fair is in games, too. On the swing, everyone gets a turn — the same number of pushes. No one is skipped. That is fair.



HERE'S THE SECRET

Fair means everyone is counted, and everyone is treated with care. It doesn't just happen by itself — we have to choose it on purpose.



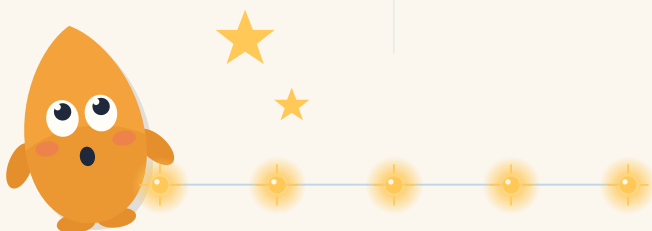
Think about it: Is “exactly the same” always fair? When might “what each one needs” be fairer?

There is a word for treating everyone with care.



FAIRNESS

You can choose this!



You do this too —



everyone gets a turn on the swing



you pick teams so no one is left out



you fix the one-street jar by adding the stones it missed

AND THE MACHINES?

Because machines learn from us, fairness must start with US. We teach it by being fair ourselves.

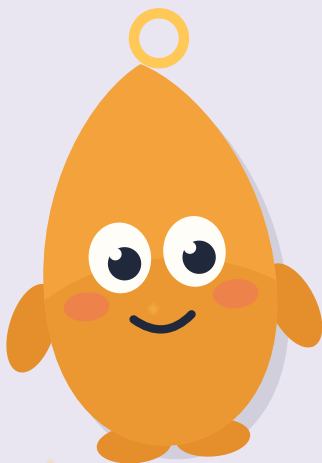
TRY IT YOURSELF

Share a snack with friends. Try “same for all” and “more for the hungriest.” Which felt fair?

FOR GROWN-UPS

Let them wrestle with it: “what would make this fair for everyone?” There’s no easy answer.

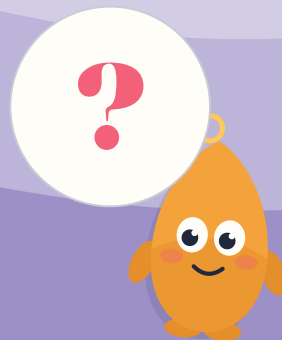
The Bird's Nest



The map tells Pixel: take the shortcut, straight through the bushes. It's the fastest way.



But right in the bushes,
Pixel sees a bird's nest with
little eggs. The fast way
would crush it. The map
doesn't know about the
nest. What should Pixel do?



LET'S FIGURE IT OUT



Pixel thinks with both head and heart. The map is right that it's faster. But the eggs are alive, and they matter more than saving a few minutes. So Pixel goes the long way around.

AND HERE'S THE TRICK

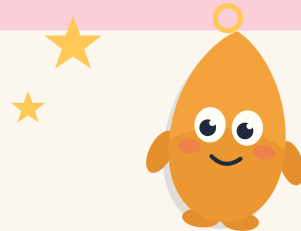


Sometimes you bend a small rule for a big, kind reason. The rule said “go straight.” But being kind to the birds was more important.



HERE'S THE SECRET

**Good choices use facts
AND care. Some choices
need a person — someone
who can feel what matters
most.**



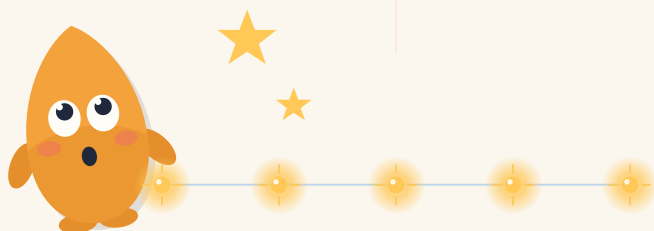
*Think about it: When should you follow the rule, and
when should you follow your heart?*

There is a word for choosing wisely, with both
facts and care.



GOOD JUDGMENT

This part is yours!



You do this too —



you pick gentle words for a sad friend



sometimes the best thing is to wait and not act



you settle a tie so everyone stays friends

AND THE MACHINES?

A machine can suggest the fast path. But deciding what really matters — that's a human job.

TRY IT YOURSELF

Find a “rule or kindness” moment today. Talk about how you would choose.

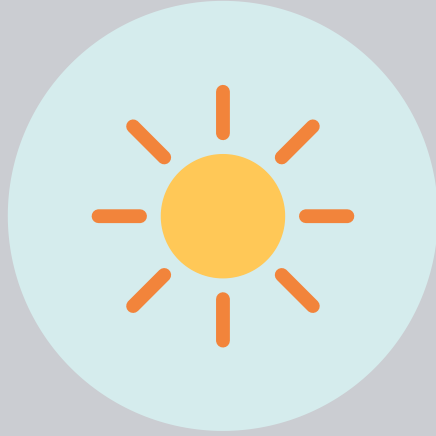
FOR GROWN-UPS

Share a time you bent a small rule for a good reason. Judgment is caught, not taught.

Should We Cross Today?



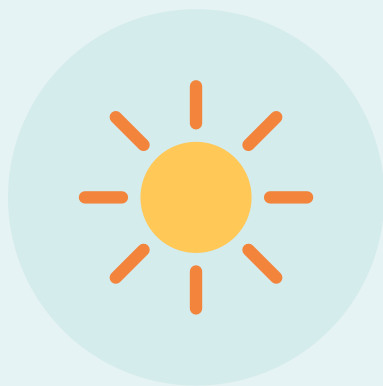
Echo knows **EVERYTHING** about the river — how deep, how fast, how cold. Echo can name every stone.



But today the river is rushing fast. Pixel asks a different question — not “what do you know?” but “should we cross today?”



LET'S FIGURE IT OUT



Echo knows all the facts. But Pixel's small question is the wise one. Knowing ABOUT the river is not the same as knowing WHEN to cross it.

AND HERE'S THE TRICK



The wisest friend isn't the one who knows the most facts. It's the one who asks the right question at the right time.



HERE'S THE SECRET

**Wisdom is not just
knowing a lot. It is
knowing what matters,
and using what you know
gently, at the right
moment.**



Think about it: Can someone know a LOT of facts and still not be wise? What might be missing?

There is a word for knowing what truly matters.



WISDOM

It grows as you grow!



You do this too —



knowing when enough is enough



choosing kind over clever



learning from an older friend's story

AND THE MACHINES?

Machines can hold a LOT of facts. But wisdom — knowing what matters — grows in caring hearts like yours.

TRY IT YOURSELF

Ask the best question you can think of today. A good question is wise, too!

FOR GROWN-UPS

Prize good questions over fast answers. “What’s worth asking here?” is a wise habit.

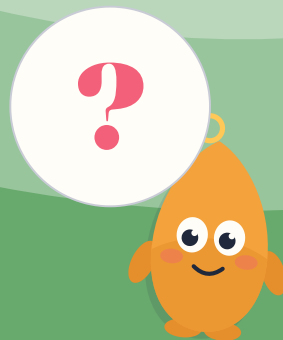
The Garden They Grew



Pixel wants to grow a giant garden. But Pixel has lots of ideas and only two little hands.



Then Pixel meets a tool that never gets tired — it can dig and water all day. But it has no ideas of its own. Can they help each other?



LET'S FIGURE IT OUT



So Pixel does the thinking —
which seeds, and where to plant
them. The tireless tool does the
digging and watering. Together,
they grow a garden neither
could grow alone!

AND HERE'S THE TRICK

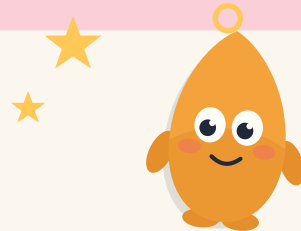


It's like a telescope. The telescope can't see FOR you — but it helps your eye see farther. The tool helps; you still do the looking.



HERE'S THE SECRET

Tools make us bigger and stronger. But people choose what to do, and aim, and care. People and tools together are best of all.



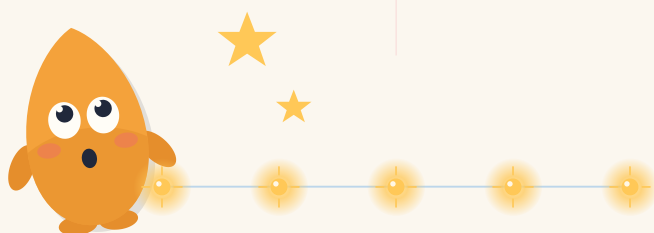
Think about it: What should stay YOUR job, even when a tool is very, very good?

There is a word for people and tools working together.



WORKING TOGETHER

You and your tools are a team!



You do this too —



a calculator adds fast; you decide what to add



a telescope helps your eye see far



a search helps you look; your good question leads

AND THE MACHINES?

The best results come from people and machines together — each doing what they're best at. That's the big idea of this whole book!

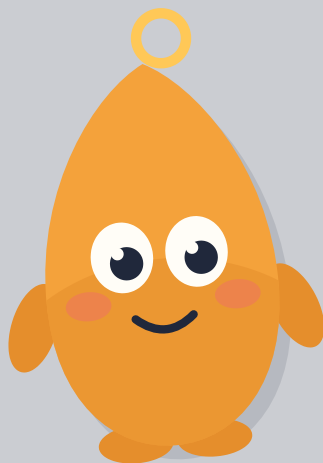
TRY IT YOURSELF

Use a tool to help make something — a ruler, a timer. You still decide what to build!

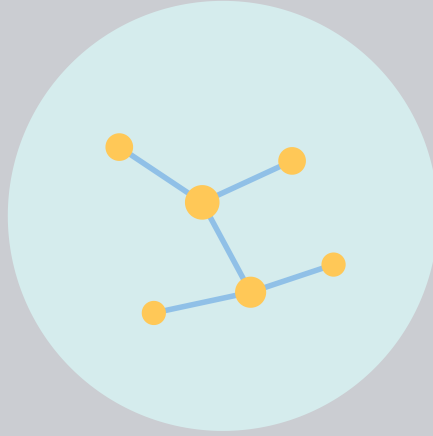
FOR GROWN-UPS

Frame tools as helpers you steer: “it can do the work; we choose what’s worth doing.”

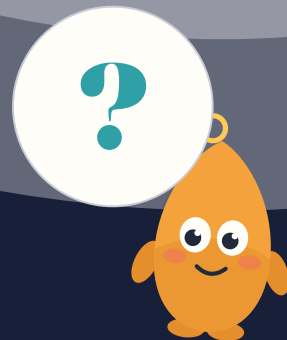
Everything Was Inside You



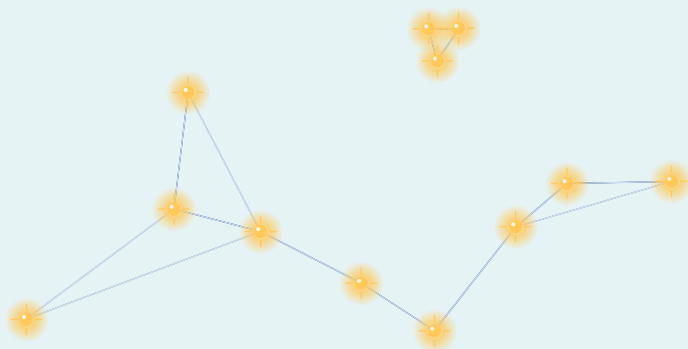
Pixel climbs back to the
hill where it all began.
Pixel looks up — and the
whole sky lights up at once.



Every star is one idea from
the journey. Guessing.
Remembering. Patterns.
Making. They are all
joined together now.
What does it mean?

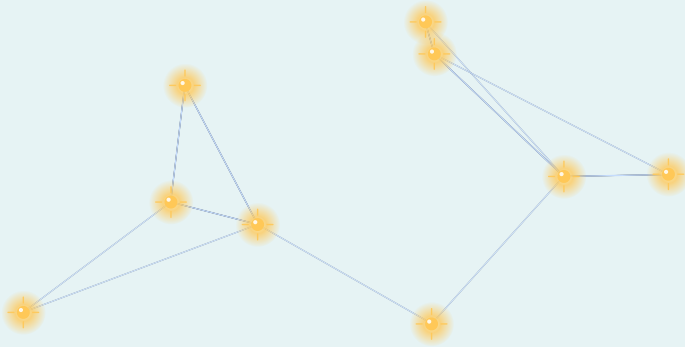


LET'S FIGURE IT OUT



Pixel sees it at last. All these ideas — predicting, learning, noticing, creating — were inside Pixel from the very start. And they are inside YOU, too.

AND HERE'S THE TRICK

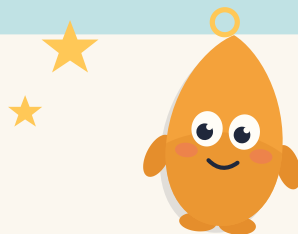


Here's the big surprise:
machines learned some of these
very skills by watching how
people think. We did not learn to
think like machines...



HERE'S THE SECRET

**Machines learned a little
from US. The word
“Artificial” was added last
— because everything else
was already inside you.**



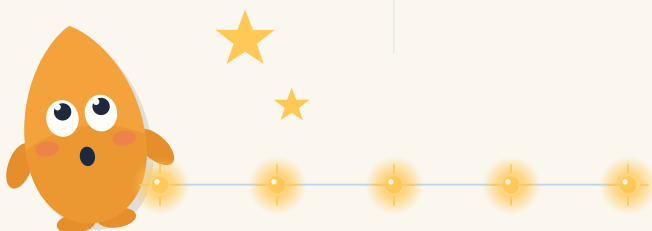
*Think about it: If all these ideas were already inside
you — what will YOU make with them?*

There is one word for all of it together.



INTELLIGENCE

And it was always yours.



You do this too —



wonder is yours



asking is yours



caring is yours

AND THE MACHINES?

Machines can do so much. But wonder, love, and wisdom stay with us. That is what makes you, *you*.

TRY IT YOURSELF

Pick your favorite idea from the whole journey.
Tell someone why it shines for you!

FOR GROWN-UPS

Close gently: “the machines learned from people — and the wonder was always yours.”

Everything Was Connected

Every idea you discovered, lit up and joined into
one sky.





The Words You Discovered

Each big word, and what it means — in a few simple words.

Prediction

a smart guess about what will come next, using clues.

Inference

figuring out something you did not see, by reading clues.

Memory

holding on to things in your mind so you can use them again.

Learning

getting to know something by seeing many examples of it.

Fixing Mistakes

noticing a mistake and changing things to get it right.

Pattern

something that repeats, so you can tell what comes next.

Features

the few important things you choose to look at.

Similarity

how close two things are, in whatever you're looking at.

Data

a big pile of examples you gather to learn from.

Training

getting good by trying, fixing, and trying again, many times.

Model

a simple copy that keeps what matters and drops the rest.

Sorting

deciding which group something belongs in, by what it's like.

Deciding

picking one thing, using what you know and what you care about.

Unfairness (Bias)

learning the wrong thing because your examples were unfair.

Network

many small parts joined up, doing something clever together.

Deep Learning

understanding big things by building up, simple to deep.

Word Machines

learning which words go together, to guess what comes next.

Making New Things

putting things you know together in a new way.

Confident Mistakes

being very sure of something that turns out not to be true.

Checking

testing something against what's real, to see if it's true.

Fairness

making sure everyone is counted and treated with care.

Good Judgment

choosing with both facts and care for what matters.

Wisdom

knowing what matters, and using what you know at the right time.

Working Together

people and tools working together, each doing what they do best.

Intelligence

all these thinking skills, working together — and they're yours.

Try It Yourself

Every little experiment from the journey, gathered here for you.

1. Prediction

Sing a song to a friend, but stop right before the last word.

2. Inference

Be a detective! Find one clue at home and guess what happened.

3. Memory

Play a memory game: look at five toys, close your eyes, and try to name them all.

4. Learning

Find five round things in your home.

5. Fixing Mistakes

Try something tricky — like stacking blocks really high.

6. Pattern

Make a clap pattern: clap-clap-stomp, clap-clap-stomp.

7. Features

Sort your toys without using color.

8. Similarity

Sort your crayons by “alike.” Then change the rule — and watch the groups change.

9. Data

Count something for a few days — like birds at a window.

10. Training

Pick one small skill.

11. Model

Draw a map of your room.

12. Sorting

Sort a drawer two different ways.

13. Deciding

Make a choice with a chart: good things on one side, not-so-good on the other.

14. Unfairness (Bias)

Ask one question to your family, then to some strangers.

15. Network

Play telephone with a long sentence.

16. Deep Learning

Build something in layers — blocks, or a drawing that starts with simple shapes.

17. Word Machines

Start a sentence and let a friend finish it.

18. Making New Things

Invent an animal from two real ones.

19. Confident Mistakes

Tell a tiny made-up “fact,” then look it up in a book.

20. Checking

Test a claim today: “this cup holds more.” Pour and find out.

21. Fairness

Share a snack with friends.

22. Good Judgment

Find a “rule or kindness” moment today.

23. Wisdom

Ask the best question you can think of today.

24. Working Together

Use a tool to help make something — a ruler, a timer.

25. Intelligence

Pick your favorite idea from the whole journey.

For Grown-Ups

READ-ALoud GUIDE

Each chapter is a little story. Read the big lines slowly, like a melody, and pause at the puzzle page so your child can guess. Let the smaller text wait for older readers.

HOW TO HELP

Ask, don't tell. The book lets a child live an idea before naming it — so try not to explain ahead. When the big word appears, enjoy the “I already knew this!” Then do the *Try It* box.

FOR TEACHERS

The five Parts build a unit: **Minds** → **Machine Learning** → **Patterns** → **Creating** → **Human Judgment**. Each chapter ends with a thought question and a hands-on activity (collected in the Appendix).

A first map of the thinking world.

Long before screens and apps, intelligence began with simple human powers — guessing, remembering, noticing patterns, learning from mistakes, imagining, and choosing what is fair. In this book, children meet each big idea as an adventure they already know, then discover its real name. By the end, they see that the ideas behind Artificial Intelligence were inside them all along.

Turns out writing a book is easier than convincing someone to publish it.

So I skipped that part.

I wrote these books for my favourite people. Since you're here, I'm assuming that's you.

They're free until a publisher discovers my existence, offers me a contract, and starts sending royalty cheques.

Enjoy the book. If you like it, drop me a note. If you love it—or if it sparks a new idea for someone you love—Hot Wheels and LEGO remain valid currency in my kingdom.

DIO.STESSO

WHAT IT BUILDS — a strong, lasting foundation of intuition for how minds and machines learn, think, and create — from first principles.

WHAT IT IS NOT — not a guide to gadgets or apps, not a coding manual, not screen time. No jargon, no hype, no scary robots — just wonder and understanding.

For absolute beginners of any age — and made for ages 6–10, who'll grow up fluent in AI. Read aloud, then read alone.